KANGHYUN RYU

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EDUCATION

University of California at Berkeley (UC Berkeley) Jan 2024 - Present (Expected May 2027)
Mechanical Engineering *Berkeley, CA*

- Ph.D. student in Intelligent Control (ICON) Lab
- M.S. in Mechanical Engineering
- Advised by Prof. Negar Mehr

University of Illinois at Urbana-Champaign (UIUC) Aug 2022 - Dec 2023
Aerospace Engineering *Champaign, IL*

- Ph.D. student in Intelligent Control (ICON) Lab; transferred to UC Berkeley

Seoul National University (SNU) Mar 2016 - Aug 2022
B.S. in Aerospace Engineering *Seoul, Korea*

- Summa Cum Laude

RESEARCH INTEREST

Robot Learning, Agentic Robotics, Multi-Agent RL, Sim2Real, Safe learning and control.

PUBLICATIONS

Google Scholar *Equal contribution

10. P. Teiletche*, **K. Ryu***, K. Park, D. Tirumala, M. Wulfmeier, and N. Mehr, "PopGen: Generating Semantically Diverse Populations for Zero-Shot Coordination", *preprint*, 2026. [Link]
9. S. Choi, S. Ye, N. Bui, A. Shrivastava, **K. Ryu**, D. Tirumala, M. Wulfmeier, and N. Mehr, "HuGo: LLMs as Whole-Body Policy Code Designers for Humanoid Loco-Manipulation", *preprint*, 2026. [Link]
8. S. Choi*, **K. Ryu***, J. Ock, and N. Mehr, "CRAFT: Coaching Reinforcement Learning Autonomously using Foundation Models for Multi-Robot Coordination Tasks", *preprint*, 2025. [Link]
7. **K. Ryu**, M. Sung, P. Gupta, J. D'sa, F. M. Tariq, D. Isele, and S. Bae, "IANN-MPPI: Interaction-Aware Neural Network-Enhanced Model Predictive Path Integral Approach for Autonomous Driving", *International Conference on Intelligent Transportation Systems (ITSC)*, 2025. [Link]
6. J.-B. Bouvier, **K. Ryu**, K. Nagpal, Q. Liao, K. Sreenath, and N. Mehr, "DDAT: Diffusion policies enforcing Dynamically Admissible robot Trajectories", *Robotics: Science and Systems (RSS)*, 2025. [Link]
5. **K. Ryu**, J.-B. Bouvier, S. Lalani, S. Eggl, and N. Mehr, "Risk-Sensitive Orbital Debris Collision Avoidance using Distributionally Robust Chance Constraints", *AIAA SCITECH Forum*, 2025. [Link]
4. **K. Ryu**, Q. Liao, Z. Li, P. Delgosha, K. Sreenath, and N. Mehr, "CurricuLLM: Automatic Task Curricula Design for Learning Complex Robot Skills using Large Language Models", *International Conference on Robotics and Automation (ICRA)*, 2025. [Link]
3. E. Clark*, **K. Ryu***, and N. Mehr, "Adaptive Learning from Demonstration in Heterogeneous Agents: Concurrent Minimization and Maximization of Surprise in Sparse Reward Environments", *Learning for Dynamics & Control Conference (L4DC)*, 2024. [Link]
2. **K. Ryu** and N. Mehr, "Integrating Predictive Motion Uncertainties with Distributionally Robust Risk-Aware Control for Safe Robot Navigation in Crowds", *International Conference on Robotics and Automation (ICRA)*, 2024. [Link]
1. **K. Ryu**, J. Kang, and D. Lee, "Performance Comparison between EKF and UKF in GPS/INS Low Observability Conditions", *The 21st International Conference on Control, Automation, and Systems (ICCAS)*, 2021. [Link]

WORK EXPERIENCES

Amazon May 2026 - Dec 2026 (ongoing)
Applied Scientist II Co-op, Industrial Robotics Group *Sunnyvale, CA*

- Agentic framework for evaluation scenario generation of high-degree of freedom robots

Honda Research Institute Jan 2025 – May 2025
PhD Research Intern *San Jose, CA*

- Developed an interaction-aware motion planning algorithm for autonomous driving in lane-changing scenario (7)
- Integrated interaction-aware trajectory prediction model with sampling-based MPPI planning algorithm

HIGHLIGHTED RESEARCH

Foundation Models for Robot Learning

- Automated task curriculum design and reward discovery for learning complex robot skills (8) (4)
- LLM-designed semantically diverse behavior sets and their corresponding policy code (10)
- LLM-designed whole-body policy code for humanoid loco-manipulation (9)

Multi-Agent Coordination

- Generation of semantically diverse agent populations for zero-shot coordination (10)
- Automated curriculum design for multi-robot coordination tasks (8)
- Adaptive teacher demonstration for heterogeneous agents by shaping student surprise (3)

Safe & Risk-Aware Control

- Diffusion policies enforcing dynamically admissible robot trajectories (6)
- LEO debris collision avoidance for satellites under distributional uncertainty in propagated trajectories (5)
- Safe navigation in human crowds coupling motion forecasting with distributionally robust control (2)

AWARDS & HONORS

Honors

- Outstanding Paper, ICRA 2026 Workshop on Scaling Compositional Intelligence for Multi-Agent Robotic Systems (8)
- Best Student Paper Finalist, selected top 6 by the Intelligent Systems Technical Committee, AIAA SciTech 2025 (5)

Fellowships

- Kwanjeong Educational Foundation Fellowship
- William C. Webster Graduate Fellowship, UC Berkeley Mechanical Engineering
- Robert Beatty Fellowship, UIUC Aerospace Engineering

Grants

- NVIDIA Academic Grant Program, 32K A100 GPU-hours (8)
- Google-BAIR Commons Program, $32K Google Cloud credits over two awards (10), (9), (6)
- UC Berkeley Conference Travel Grant, L4DC 2024 (3)
- IEEE Robotics and Automation Society Travel Grant, ICRA 2024 (2)

TEACHING

ME 132 Dynamic Systems and Feedback - Graduate Teaching Assistant Fall 2025

SKILLS

Programming Language : Python, C++, MATLAB, Julia
Software & Tools : Isaac Lab, mjlab, MuJoCo, ROS, Strands Agents, Unitree G1, Unitree Go2, Motion capture

SERVICES

Conference Reviewer : ICRA 2026, 2027; IROS 2025, 2026; CoRL 2026; L4DC 2025; CDC 2023, 2024, 2026; ITSC 2025; WAFR 2024; ISRR 2024
Journal Reviewer : T-RO, TCST, RA-L